Selfrace: A Reinforcement Learning Framework for Autonomous Driving Simulation

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Abstract

This file is a ready-to-build Elsevier journal manuscript template based on the `elsarticle` class. Replace this paragraph with a concise summary of the research problem, method, main results, and implications.

Keywords: autonomous driving, reinforcement learning, simulation, decision making

1. Introduction

Autonomous driving research often depends on controllable simulation environments for training and evaluation. This manuscript template can be used as a starting point for a journal submission and already includes common elements such as citations, equations, figures, and tables.

The source code for this project can be cited as supporting implementation material. Replace this sentence with the contribution statement of the paper.

2. Related Work

Journal manuscripts usually summarize the closest lines of work and explain how the proposed method differs from them. A citation example is shown here [1].

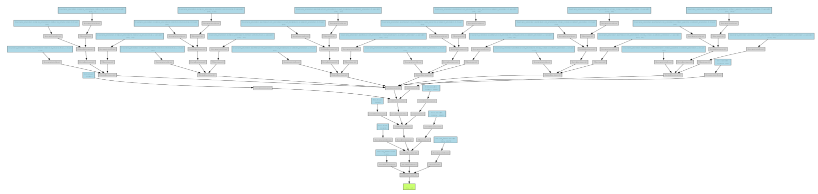

Figure 1: Example figure placeholder. Replace it with the architecture or experiment diagram for the manuscript.

Table 1: Example experimental results. Replace the values with real metrics.

Method	Success Rate	Collision Rate	Mean Reward
Baseline	0.72	0.18	124.3
Proposed	0.86	0.09	168.7

3. Method

Let s_t denote the simulation state, a_t the selected action, and r_t the reward at time step t. A standard objective is

$$J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T} \gamma^t r_t\right], \tag{1}$$

where γ is the discount factor and π_θ is the policy.

4. Experiments

Describe datasets, scenarios, baselines, metrics, hardware, and random seeds. Table 1 shows a compact result-table style.

5. Conclusion

Summarize the main findings, limitations, and future work. Keep this section short and specific.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

The code and data availability statement should be updated before submission.

References

[1] R. S. Sutton, A. G. Barto, Reinforcement Learning: An Introduction, 2nd Edition, MIT Press, 2018.